

DevOps Lab Programs

Program7: Deploy a web application to kubernetes using Minikube

Structure of the Program

→nginx-deployment.yaml

Steps for execution:

Step 1:Start the minikube

```
minikube start
```

Step 2: Check the status of the minikube

```
minikube status
```

Step 3: if some services didn't start then run the following command

```
minikube delete
```

```
minikube start --driver=docker
```

step 4: Build the deployment yaml file



```
GNU nano 8.1
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 5: Apply the nginx-deployment.yaml file

```
kubectl apply -f nginx-deployment.yaml
```

Step 6: Check the Status

```
kubectl get pods
```

```
kubectl get deployments
```

Step 7: Expose the deployment as service

Step 8: Check the services

Step 9: Access the web application

The screenshot shows a terminal window with the following commands and output:

```
minikube addons enable metrics-server
minikube
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

1rv24nc083-rakshithadas@1rv24nc083rakshithadas:~/Docker/Lab7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment configured
1rv24nc083-rakshithadas@1rv24nc083rakshithadas:~/Docker/Lab7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 2/2     2            2           19h
1rv24nc083-rakshithadas@1rv24nc083rakshithadas:~/Docker/Lab7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5lgf7  1/1     Running   2 (66s ago)  33m
nginx-deployment-6f9664446b-d5nqw  1/1     Running   2 (66s ago)  33m
1rv24nc083-rakshithadas@1rv24nc083rakshithadas:~/Docker/Lab7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
1rv24nc083-rakshithadas@1rv24nc083rakshithadas:~/Docker/Lab7$ kubectl get services
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes    ClusterIP   10.96.0.1     <none>         443/TCP   23d
kubia         NodePort    10.106.174.127 <none>         8080:32481/TCP 23d
nginx-deployment NodePort    10.118.47.236 <none>         80:30331/TCP 11s
nginx-service NodePort    10.97.155.196 <none>         80:30866/TCP 19h
1rv24nc083-rakshithadas@1rv24nc083rakshithadas:~/Docker/Lab7$ minikube service nginx-deployment

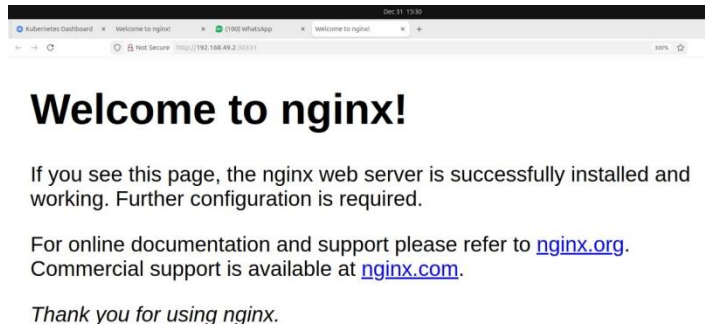
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30331

Opening service default/nginx-deployment in default browser...

1rv24nc083-rakshithadas@1rv24nc083rakshithadas:~/Docker/Lab7\$ Gtk-Message: 15:30:02.400: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.

Step 10: View the output in the web browser



Step 11: Delete all the pods

Step 12: Delete all the services

Step 13: Delete all the deployments